

Brac University

Department of Electrical & Electronic Engineering

Semester Spring-25

Course Number: EEE103L

Course Title: Computer Programming Laboratory

Section: 06

Lab Report



Experiment no.

Name of the experiment: Arrays and string in C language

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Lab report of experiment 5

a. Write a program in C using an array which finds the largest and smallest elements in a group of numbers.

Code:

```
#include <stdio.h>

int main()
{
    int n, i;

    int numbers[100];

    int max, min;

    printf("Enter the number of elements: ");

    scanf("%d", &n);

    printf("Enter %d numbers:\n", n);

    for (i = 0; i < n; i++)

        scanf("%d", &numbers[i]);

    max = min = numbers[0];

    for (i = 1; i < n; i++) {

        if (numbers[i] > max) {

            max = numbers[i];

        }

        if (numbers[i] < min) {

            min = numbers[i];

        }

    }

    printf("Largest number: %d\n", max);

    printf("Smallest number: %d\n", min);
```

```
return 0;
```

```
}
```

Screenshot of code:

```
D: > BracU > 4. Fourth Semester (SP-25) > EEE103 > Lab report > exp 5 > C high low number.c >
1  #include <stdio.h>
2
3  int main()
4  {
5      int n, i;
6      int numbers[100];
7      int max, min;
8
9
10     printf("Enter the number of elements: ");
11     scanf("%d", &n);
12
13     printf("Enter %d numbers:\n", n);
14     for (i = 0; i < n; i++)
15         scanf("%d", &numbers[i]);
16
17     max = min = numbers[0];
18
19     for (i = 1; i < n; i++) {
20         if (numbers[i] > max) {
21             max = numbers[i];
22         }
23         if (numbers[i] < min) {
24             min = numbers[i];
25         }
26     }
27
28     printf("Largest number: %d\n", max);
29     printf("Smallest number: %d\n", min);
30
31     return 0;
32 }
33
```

Output:

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS C:\Users\aryan> & 'c:\Users\aryan\.vscode\extensions\ms-vscode.cmake-tools\bin\cmake\bin\cmake.exe' -G 'Visual Studio 15' -S . -B build -D CMAKE_BUILD_TYPE=Debug -D CMAKE_C_COMPILER='c:\Users\aryan\.vscode\extensions\ms-vscode.cmake-tools\bin\cmake\bin\gcc.exe' -D CMAKE_CXX_COMPILER='c:\Users\aryan\.vscode\extensions\ms-vscode.cmake-tools\bin\cmake\bin\g++.exe' --stdout=Microsoft-MIEngine-Pid-kyqsspt5.azr' --dbgExe=C:\msys64\bin\gdb.exe
Enter the number of elements: 4
Enter 4 numbers:
21
56
45
90
Largest number: 90
Smallest number: 21
```

b. Write a program in C to augment one string with another string. For example: String 'a' contains "foot", string 'b' contains "ball", now string 'c' should contain "football".

Code:

```
#include <stdio.h>

#include <string.h>

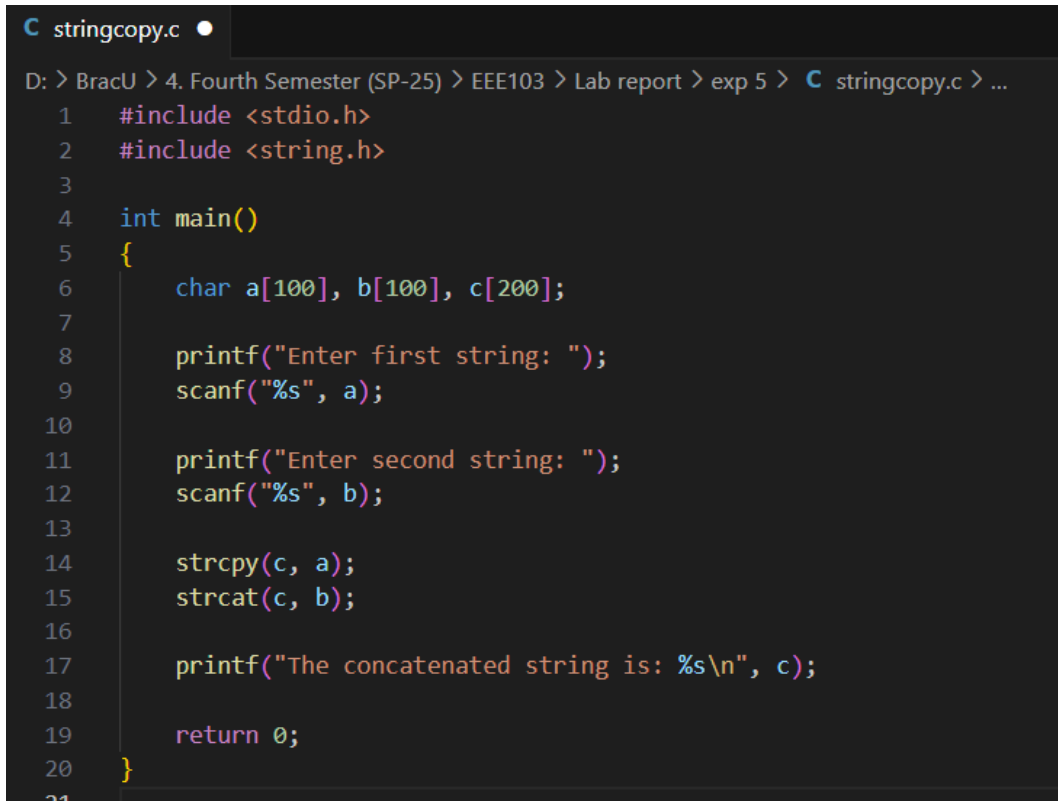
int main()
{
    char a[100], b[100], c[200];
    printf("Enter first string: ");
    scanf("%s", a);
    printf("Enter second string: ");
    scanf("%s", b);
    strcpy(c, a);
    strcat(c, b);
```

```
printf("The concatenated string is: %s\n", c);
```

```
return 0;
```

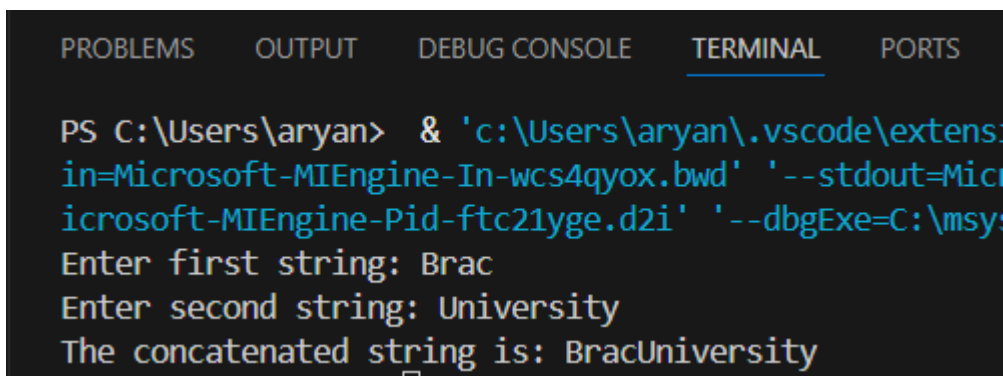
```
}
```

Screenshot of code:



```
C stringcopy.c
D: > BracU > 4. Fourth Semester (SP-25) > EEE103 > Lab report > exp 5 > C stringcopy.c > ...
1  #include <stdio.h>
2  #include <string.h>
3
4  int main()
5  {
6      char a[100], b[100], c[200];
7
8      printf("Enter first string: ");
9      scanf("%s", a);
10
11     printf("Enter second string: ");
12     scanf("%s", b);
13
14     strcpy(c, a);
15     strcat(c, b);
16
17     printf("The concatenated string is: %s\n", c);
18
19     return 0;
20 }
```

Output:



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
PS C:\Users\aryan> & 'c:\Users\aryan\.vscode\extensions\Microsoft-MIEngine-In-wcs4qyox.bwd' '--stdout=Microsoft-MIEngine-Pid-ftc21yge.d2i' '--dbgExe=C:\msys64\bin\cmd.exe'
Enter first string: Brac
Enter second string: University
The concatenated string is: BracUniversity
```

c. Write a program to print a string in reverse order

Code:

```
#include <stdio.h>

#include <string.h>

int main()

{

char str[200];

int i, length;

printf("Enter a string: ");

fgets(str, sizeof(str), stdin);

length = strlen(str);

if (str[length - 1] == '\n')

{

str[length - 1] = '\0';

length--;

}

printf("Reversed string: ");

for (i = length - 1; i >= 0; i--)

printf("%c", str[i]);

printf("\n");

return 0;

}
```

Screenshot of code:

```
C reverse.c
D: > BracU > 4. Fourth Semester (SP-25) > EEE103 > Lab report > exp 5 > C r
1  #include <stdio.h>
2  #include <string.h>
3
4  int main()
5  {
6      char str[200];
7      int i, length;
8
9      printf("Enter a string: ");
10     fgets(str, sizeof(str), stdin);
11
12     length = strlen(str);
13     if (str[length - 1] == '\n')
14     {
15         str[length - 1] = '\0';
16         length--;
17     }
18
19     printf("Reversed string: ");
20     for (i = length - 1; i >= 0; i--)
21         printf("%c", str[i]);
22
23     printf("\n");
24
25     return 0;
26 }
27
```

Output:

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL
PS C:\Users\aryan> & 'c:\Users\aryan\.vscode
in=Microsoft-MIEngine-In-0mbwayoc.xej' '--st
icrosoft-MIEngine-Pid-mmszm1ws.hkf' '--dbgEx
Enter a string: BRAC UNIVERSITY
Reversed string: YTISREVINU CARB
```

d. A palindrome is a word, number or other sequence of characters which reads the same backward as forward, such as 'madam'. Write a C program which can determine if a word is palindrome or not.

Code:

```
#include <stdio.h>

#include <string.h>

#include <ctype.h>

int main() {

char str[200], cleaned[200];

int i, j = 0, length, isPalindrome = 1;

printf("Enter a string: ");

fgets(str, sizeof(str), stdin);

length = strlen(str);

if (str[length - 1] == '\n') {

str[length - 1] = '\0';

length--;

}

for (i = 0; i < length; i++) {

if (isalnum(str[i])) {

cleaned[j++] = tolower(str[i]);

}

}

cleaned[j] = '\0';

for (i = 0; i < j / 2; i++) {

if (cleaned[i] != cleaned[j - 1 - i]) {
```



```
isPalindrome = 0;
```

```
break;
```

```
}
```

```
}
```

```
if (isPalindrome)
```

```
printf("The text is a palindrome.:D \n");
```

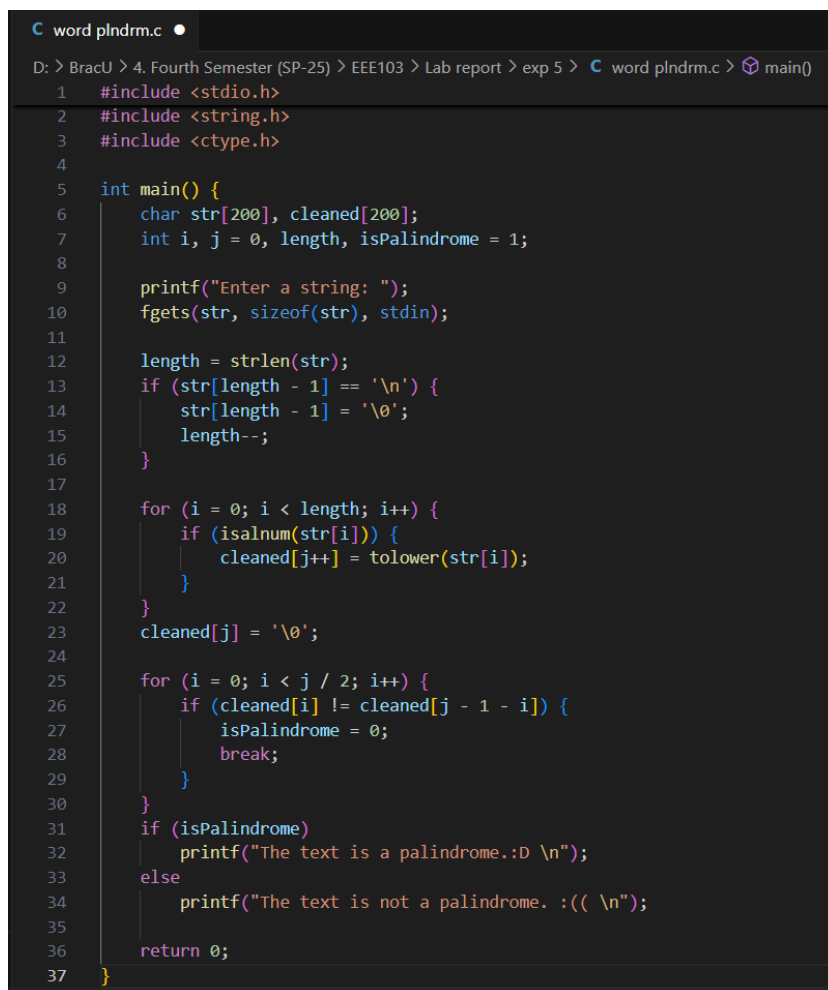
```
else
```

```
printf("The text is not a palindrome. :(( \n");
```

```
return 0;
```

```
}
```

Screenshot of code:

A screenshot of a code editor window titled 'word plndrm.c'. The editor shows a C program that checks if a string is a palindrome. The code includes standard headers, a main function that reads a string, cleans it (removes non-alphanumeric characters and converts to lowercase), and then compares characters from both ends to determine if it's a palindrome. The output messages use emoticons to indicate success or failure.

```
C word plndrm.c
D: > BracU > 4. Fourth Semester (SP-25) > EEE103 > Lab report > exp 5 > C word plndrm.c > main()
1  #include <stdio.h>
2  #include <string.h>
3  #include <ctype.h>
4
5  int main() {
6      char str[200], cleaned[200];
7      int i, j = 0, length, isPalindrome = 1;
8
9      printf("Enter a string: ");
10     fgets(str, sizeof(str), stdin);
11
12     length = strlen(str);
13     if (str[length - 1] == '\n') {
14         str[length - 1] = '\0';
15         length--;
16     }
17
18     for (i = 0; i < length; i++) {
19         if (isalnum(str[i])) {
20             cleaned[j++] = tolower(str[i]);
21         }
22     }
23     cleaned[j] = '\0';
24
25     for (i = 0; i < j / 2; i++) {
26         if (cleaned[i] != cleaned[j - 1 - i]) {
27             isPalindrome = 0;
28             break;
29         }
30     }
31     if (isPalindrome)
32         printf("The text is a palindrome.:D \n");
33     else
34         printf("The text is not a palindrome. :(( \n");
35     return 0;
36 }
37 }
```

Output:

Not a palindrome

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\aryan> & 'c:\Users\aryan\.vscode\extensions\Microsoft-MIEngine-In-pr2j4dcb.ftr' '--stdout=Microsoft-MIEngine-Pid-gdfa5qto.h0v' '--dbgExe=C:\msys64\bin\cmd.exe'
Enter a string: Dept of EEE BracU
The text is not a palindrome. :((
```

Palindrome sentence

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\aryan> & 'c:\Users\aryan\.vscode\extensions\Microsoft-MIEngine-In-p1xhk40z.sdw' '--stdout=Microsoft-MIEngine-Pid-okvzebtp.4zf' '--dbgExe=C:\msys64\bin\cmd.exe'
Enter a string: mr owl ate my metal worm
The text is a palindrome.:D
```